

A Consulting System for Portfolio Assets Allocation by Selecting the Best Agent in the Short Term Based on Cumulative Returns with Deep Reinforcement Learning

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Abstract—This paper proposes a sophisticated consulting system designed to optimize investment portfolios using deep reinforcement learning (DRL). The system includes a unique decision-making mechanism that selects the best-performing agent in the short term based on cumulative returns. We employ five different DRL agents: Advantage Actor-Critic (A2C), Soft Actor-Critic (SAC), Twin Delayed Deep Deterministic Policy Gradient (TD3), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO), to allocate portfolio weights dynamically. By evaluating each agent's performance over the previous ten days, our system adaptively selects the agent most likely to maximize returns in the immediate future. This approach enhances portfolio performance by leveraging the strengths of multiple DRL agents. The effectiveness of our proposed method is validated using historical data from the Dow Jones index, demonstrating significant improvements in cumulative returns and risk-adjusted performance metrics. Our results show that the main proposed method achieved the highest annual return of 11.43%, cumulative returns of 38.29%, and a Sharpe ratio of 0.832, outperforming individual DRL agents and highlighting the potential of DRL in advancing portfolio management strategies.

Keywords—Deep Reinforcement Learning, Consulting System, Portfolio Optimization, Tactical Decision, Cumulative Return

I. INTRODUCTION

In the ever-evolving landscape of financial markets, effective portfolio management is crucial for investors aiming to maximize returns while minimizing risks. Traditional portfolio optimization techniques often rely on static models, which can be inadequate in capturing the dynamic nature of financial markets [1]. Sequence processing techniques, such as linear vectors, offer a simple and efficient way to handle financial data. These approaches are well-suited for portfolio optimization tasks where low complexity is desirable, providing a straightforward method for analyzing data. Linear vectors, along with vectorization and sampling, can be applied effectively to time series data, such as financial data, portfolios, cryptocurrency, and signals, where patterns play a critical role over time [2]. In this context, electronic consulting plays a crucial role in distilling complex financial data into actionable insights for investors. With the rapid advancement of technology, having access to sophisticated analytical tools is essential for navigating the complexities of modern financial markets [3]. To address this challenge, we propose a consulting system that leverages deep reinforcement learning (DRL) to enhance portfolio asset allocation.

Deep reinforcement learning has shown significant promise in various domains due to its ability to learn and adapt to complex environments. In financial markets, DRL can be

particularly powerful, enabling adaptive decision-making based on real-time data and historical trends [4]. Our proposed system integrates multiple reinforcement learning agents to dynamically allocate weights to stocks in a portfolio, thereby optimizing asset allocation.

One of the unique features of our system is its ability to select the best-performing agent over a short-term period based on cumulative returns. By evaluating the performance of each agent over the previous ten days, the system dynamically chooses the agent that is most likely to maximize returns in the immediate future. This short-term selection mechanism allows the system to adapt quickly to changing market conditions, enhancing its overall performance.

In our methodology, we employ five reinforcement learning agents: Advantage Actor-Critic (A2C), Soft Actor-Critic (SAC), Twin Delayed Deep Deterministic Policy Gradient (TD3), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO). Each agent brings its strengths and approaches to the table, and by integrating their predictions, our system aims to achieve superior portfolio performance.

The contributions of this paper are threefold:

- We propose a novel consulting system for portfolio asset allocation that leverages deep reinforcement learning.
- We introduce a short-term agent selection mechanism based on cumulative returns to enhance adaptive decision-making.
- We validate the effectiveness of our approach using historical financial data from the Dow Jones index.

The remainder of this paper is structured as follows. Section 2 reviews related works in portfolio optimization and reinforcement learning. Section 3 details our methodology, including data preprocessing, model training, and the agent selection process. Section 4 presents the experimental results and performance evaluation. Section 5 is about discussion. Finally, section 6 concludes the paper and discusses potential future research directions.

II. RELATED WORK

Recent advancements in portfolio management have seen a surge in the application of innovative reinforcement learning (RL) architectures. Betancourt and Chen (2021) [5] addressed markets with dynamic assets by developing a method that integrates new assets without retraining, which yielded impressive daily returns. Katongo and Bhattacharyya (2021) [6] employed deep reinforcement learning algorithms and

neural network autoencoders for tactical asset allocation. Their approach, applied to the Dow Jones Industrial Average, aimed to optimize portfolio performance by increasing risk-adjusted returns. Koratamaddi et al. (2021) [7] introduced a novel RL approach incorporating market sentiment to train an automated trader. The model, which used both historical stock prices and market sentiment data, demonstrated superior performance across metrics like the Sharpe ratio and annualized returns compared to existing baselines. Lim, Cao, and Quek (2022) [8] enhanced dynamic portfolio rebalancing by combining an RL agent with the Long Short-Term Memory (LSTM) model, resulting in significant return improvements. Similarly, Kabbani and Duman (2022) [9] applied the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm to automate stock trading, achieving a high Sharpe ratio. Zhao et al. (2023) [10] utilized a self-attention mechanism to address complex asset correlations, presenting a deterministic policy gradient recurrent RL method that performed excellently across various financial datasets. Li et al. (2023) [11] utilized a framework called LSRE-CAAN with components like a long sequence representation extractor and cross-asset attention network, achieving impressive returns and lower risk metrics using high-frequency cryptocurrency data. Ma et al. (2023) [12] proposed a multi-agent system with trend consistency regularization to dynamically switch between agents based on market status, showing effectiveness in the Chinese Stock Market. Hao et al. (2023) [13] combined fuzzy vectors with multiple RL algorithms, applying an ensemble method to SP100 stocks, and demonstrated better performance than benchmarks over 11 years.

In contrast, other studies have focused on integrating RL with additional financial theories or methods to further refine portfolio management strategies. Zhang et al. (2020) [14] proposed a cost-sensitive portfolio selection method that managed transaction and risk costs effectively through a two-stream portfolio policy network. Soleymani and Paquet (2020) [15] developed the DeepBreath framework, which integrates a restricted stacked autoencoder and convolutional neural network (CNN) for dimensionality reduction and policy enforcement, respectively, and also utilized blockchain technology to solve settlement issues. Wu et al. (2020) [16] employed the Gated Recurrent Unit (GRU) to design adaptive stock trading strategies, surpassing traditional methods like the Turtle trading strategy. Wu et al. (2021) [17] created a system using CNN and RNN models with a Sharpe ratio-based reward function, showing significant improvements in returns and reduced drawdown risks, particularly effective in traditional stock datasets. Carta et al. (2021) [18] utilized multiple deep neural networks and a reward-based classifier to maximize profit, demonstrating superior performance on the S&P 500 and individual stocks like J.P. Morgan and Microsoft compared to baseline and buy-and-hold strategies. Lin et al. (2022) [19] developed a multi-agent RL framework with a nested agent structure and specially designed reward functions to manage risk shifting in portfolio management, outperforming traditional portfolio selection strategies. Jang and Seong (2023) [20] combined modern portfolio theory with deep reinforcement learning, using Tucker decomposition to optimize multimodal inputs and outperform state-of-the-art algorithms. Day et al. (2023) [21] incorporated technical indicators, returns, and covariance in a dynamic trading strategy model based on RL, achieving high annualized returns and cumulative returns on the iShares MSCI US ESG Select Index ETF. Yu et al. (2023) [22] introduced a nested

RL method that integrated several decision-making models and a weight random selection strategy to enhance investor profits across various market environments, outperforming baseline strategies in the U.S., Japanese, and British stock markets. Li and Hai (2024) [23] introduced a multi-agent deep reinforcement learning model that incorporated additional stock financial indices and self-attention networks, significantly improving portfolio management and risk mitigation.

III. METHODS

This section describes the development of a portfolio asset allocation consulting system using five reinforcement learning agents (A2C, PPO, DDPG, TD3, SAC) to optimize daily stock weights for maximizing cumulative returns through deep reinforcement learning. Each agent, including A2C with its dual neural network approach for policy and value determination, PPO with synchronous policy updates, DDPG with target networks and replay buffer for stability, TD3 with synchronous value updates, and SAC with probabilistic policy for exploration, offers distinct strengths suited to varying market conditions. Combining these agents leverages their advantages in predicting dynamic and complex financial markets, selecting the most profitable agent based on its recent performance to forecast the current day's market trends.

In our approach, raw data comprising 4,035 working days and five columns of opening price, closing price, maximum price, minimum price, and trading volume for each day for 30 stocks is first downloaded ($s \times k \times 5$). Using the closing price ($s \times k \times 1$) for each working day, 20 stocks with the least volatility are then selected using an autoencoder, resulting in stocks with less deviation from the average and lower risk ($s \times k' \times 1$). Next, 10 indicators are applied to the stocks to predict trends ($s \times k \times 10$), and a neural network autoencoder reduces the number of these indicators to four to optimize the training process ($s \times k \times 4$). The dataset ($s \times k \times 5$), the reduced number of stocks ($s \times k' \times 1$), and the applied indicators ($s \times k \times 4$) are then aggregated ($s \times k' \times 9$). The following steps outline our comprehensive approach.

A. Stock Selection

Initially, 30 stocks from the Dow Jones are selected as portfolio elements. An autoencoder neural network is employed to select 20 stocks with the lowest volatility, aiming to reduce risk by filtering out stocks with high volatility. This selection process ensures that the portfolio consists of relatively stable stocks.

B. Indicator Application

Various technical indicators are applied to each stock's data to predict trends and understand market behavior. These indicators include Average True Range (ATR), Bollinger Bands Width (BBW), On-Balance Volume (OBV), Chaikin Money Flow (CMF), Moving Average Convergence Divergence (MACD), Average Directional Index (ADX), Relative Strength Index (RSI), Commodity Channel Index (CCI), Exponential Moving Average (EMA), and Simple Moving Average (SMA). Applying these indicators helps in capturing different aspects of market trends and stock performance.

C. Dimensionality Reduction

To optimize the learning process, the dimensions of the indicators are reduced from 10 to 4 using an autoencoder

neural network. This reduction helps in managing computational complexity and enhances the efficiency of the learning algorithms by focusing on the most relevant features of the indicators.

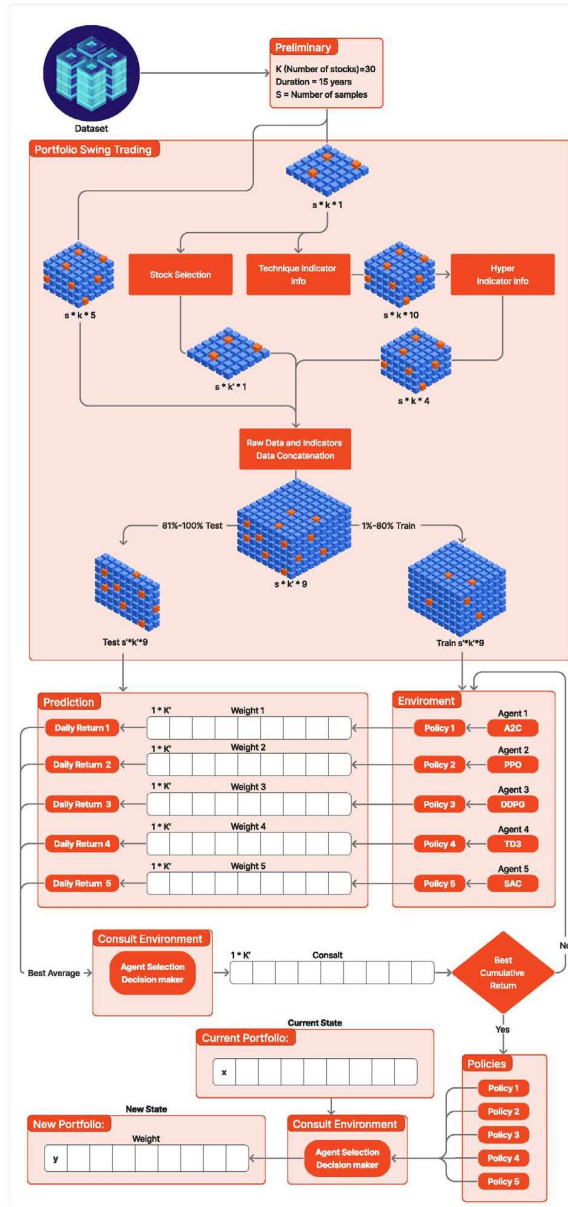


Fig. 1. General schema of our proposed consulting system.

D. Raw Data and Indicators Concatenation

The raw data, consisting of the close price, open price, low price, high price, and volume for each stock, is concatenated with the reduced indicator data. This combined dataset provides a comprehensive view of each stock's performance and market behavior, crucial for the subsequent training process. Specifically, the values of the lowest price, the highest price, the closing price, the opening price, the trading volume, and four indicators for the 20 selected stocks are included.

E. Data Division

The data is divided into training and testing sets, ensuring that the models are well-validated and can generalize to unseen data. Typically, 80% of the data is used for training the models, and the remaining 20% is used for testing. Given the time-series nature of the data, this split is not random to preserve temporal dependencies.

F. Training the DRL Agents

Five reinforcement learning agents (A2C, PPO, DDPG, TD3, SAC) are trained on the dataset. Each agent uses specific hyperparameters tailored to their algorithms to learn and predict the optimal weights for the stocks. Agents are trained to maximize daily income and minimize the Sharpe ratio.

G. Weight Allocation Using RL Agents

The portfolio consists of 20 stocks, each with a specific weight indicating its proportion within the portfolio. After being trained, the agents predict these weights for each working day during the backtesting phase to maximize daily income and minimize the Sharpe ratio.

H. Consulting System

The consulting system uses the previous ten days' outputs from the five agents to predict the next day's returns. By examining the daily average income of all agents over the preceding days, the weight of the best-performing agent in the previous ten days is used for stock weighting on the current day. This algorithm sums the total daily income of all agents, divides each agent's income by this total, and determines each agent's contribution to the ten-day income. The highest-performing agent over the past ten days is selected as the chosen agent for the current day, potentially surpassing the cumulative income of the highest-performing agent in some cases.

I. Hyperparameters

Hyperparameters play a crucial role in shaping the learning behavior and performance of each reinforcement learning agent. These parameters are carefully tuned to enhance the agents' ability to adapt and optimize decision-making processes within the portfolio optimization context. Below are the key hyperparameters used for each agent in our proposed method:

- 1) **Total_timesteps**: This parameter determines the duration of model training across all agents, set to 10,000 timesteps.
- 2) **Learning_rate**: A critical parameter for adjusting the pace of learning, standardized to 0.001 across all agents.
- 3) **N_steps**: For A2C and PPO, this signifies the number of steps used for updating advantages, set to 5.
- 4) **Ent_coef**: Used by A2C and PPO to represent the balance between exploration and exploitation, set to 0.005.
- 5) **Batch_size**: For PPO, DDPG, TD3, and SAC, this sets the number of samples used for model updates, specified as 1024.
- 6) **Buffer_size**: Applicable to DDPG, TD3, and SAC, indicating the size of the replay buffer which stores past experiences for learning, set to 50,000.

- 7) **Learning starts:** Unique to SAC, defining the number of initial steps before learning begins, specified as 100.

These hyperparameters collectively shape the learning dynamics of the agents, enabling fine-tuning to achieve a balance between the exploration of new strategies and the exploitation of known profitable strategies.

J. Aggregation and Evaluation

In our proposed method, the aggregation and evaluation process is critical for determining the effectiveness of combining multiple reinforcement learning agents for portfolio asset allocation. This section describes the steps involved in aggregating the agents' predictions and evaluating their performance. The evaluation methods we used are:

- 1) **Data Division:** The dataset is divided into training and testing sets, with 80% of the data used for training the models and the remaining 20% for testing. This time-series data split ensures the models are validated effectively and can generalize to unseen data.
- 2) **Performance Metrics:** Several key performance metrics are employed to evaluate the effectiveness of the proposed method:
 - **Annual Return:** The return achieved over a year.

$$\text{Annual Return} = \left(\frac{\text{End Value}}{\text{Start Value}} \right)^{\frac{1}{n}} - 1 \quad (1)$$

Where n is the number of years.

- **Cumulative Returns:** The total return accumulated over the entire evaluation period.

$$\text{Cumulative Returns} = \frac{\text{End Value} - \text{Start Value}}{\text{Start Value}} \quad (2)$$

- **Annual Volatility:** The standard deviation of returns over a year.

$$\text{Annual Volatility} = \sigma \times \sqrt{252} \quad (3)$$

Where σ is the standard deviation of daily returns and 252 represents the average number of trading days in a year.

- **Average Daily Returns:** The mean return achieved daily.

$$\text{Average Daily Returns} = \frac{1}{n} \sum_{i=1}^n R_i \quad (4)$$

Where n is the number of trading days, and R_i is the return on day i .

- **Sharpe Ratio:** A measure of risk-adjusted return, calculated as the ratio of average daily returns to the standard deviation of returns.

$$\text{Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p} \quad (5)$$

Where $E[R_p]$ represents the expected return of the portfolio, R_f denotes the risk-free rate, and σ_p stands for the standard deviation of the portfolio's excess returns.

- **Calmar Ratio:** The annual return divided by the maximum drawdown.

$$\text{Calmar Ratio} = \frac{\text{Annual Return}}{\text{Max drawdown}} \quad (6)$$

- **Stability:** This metric is usually defined in the context of time series and reflects the consistency of returns, often assessed using statistical tests or measures such as the Hurst exponent.
- **Max Drawdown:** The maximum observed loss from a peak to a trough.

$$\text{Max Drawdown} = \min_t \left(\frac{\text{Trough}_t - \text{Peak}_t}{\text{Peak}_t} \right) \quad (7)$$

Where Peak_t and Trough_t are the highest and lowest points in the portfolio's value before a new peak is achieved.

- **Omega Ratio:** The ratio of the probability-weighted gains to the probability-weighted losses.

$$\text{Omega Ratio} = \frac{\int_{R_f}^{\infty} [1 - F(x)] dx}{\int_{-\infty}^{R_f} F(x) dx} \quad (8)$$

Where $F(x)$ is the cumulative distribution function of returns, and R_f is the threshold return (often the risk-free rate).

- **Sortino Ratio:** An adjusted version of the Sharpe ratio that distinguishes between downside risk and overall volatility by focusing on the standard deviation of negative asset returns.

$$\text{Sortino Ratio} = \frac{E[R_p] - R_f}{\sigma_d} \quad (9)$$

Where σ_d is the standard deviation of negative asset returns.

- **Skew:** The measure of the asymmetry of the return distribution.

$$\text{Skew} = \frac{E[(R_p - \mu)^3]}{\sigma^3} \quad (10)$$

Where μ is the mean return and σ is the standard deviation of returns.

- **Kurtosis:** The measure of the "tailedness" of the return distribution.

$$\text{Kurtosis} = \frac{E[(R_p - \mu)^4]}{\sigma^4} \quad (11)$$

- **Tail Ratio:** The ratio of the average of the upper 5% of returns to the average of the lower 5%.

$$\text{Tail Ratio} = \frac{\text{Average of Top 5\% Returns}}{\text{Average of Bottom 5\% Returns}} \quad (12)$$

- **Daily Value at Risk (VaR):** The maximum loss expected (with a given confidence level) over 24 hours.

$$\text{VaR}_\alpha = -\inf\{x | F(x) \geq \alpha\} \quad (13)$$

Where $F(x)$ is the cumulative distribution function of returns, and α is the confidence level (e.g., 0.05 for 95% confidence).

- **Alpha:** A metric that quantifies the excess return on an investment relative to the benchmark market index.

$$\alpha = R_p - [R_f + \beta(R_m - R_f)] \quad (14)$$

Where R_p is the portfolio return, R_f is the risk-free rate, β is the portfolio's beta, and R_m is the market return.

- **Beta:** A metric that assesses the volatility or systematic risk of a portfolio in comparison to the overall market.

$$\beta = \frac{Cov(R_p, R_m)}{Var(R_m)} \quad (15)$$

Where $Cov(R_p, R_m)$ is the covariance of the portfolio and market returns, and $Var(R_m)$ is the variance of the market returns.

- 3) **Comparative Analysis:** The performance of the proposed method is compared against individual agent performance and traditional portfolio allocation strategies. This comparative analysis highlights the benefits of using a combined agent approach over-relying on a single agent or conventional methods.
- 4) **Robustness Check:** Additional tests are conducted to ensure the robustness of the method under different market conditions and data subsets. This involves stress-testing the model with various market scenarios to verify its stability and performance consistency.

IV. RESULTS

In this section, we present the results of our experiments, beginning with an overview of the dataset utilized and followed by an analysis of the performance of individual reinforcement learning agents and a comparative evaluation of different strategies and algorithms.

A. Dataset Description

We utilize the FinRL library [24] to download the dataset, sourced from the Yahoo Finance website's API, where data related to the Dow Jones index is freely available. The Dow Jones Industrial Average (DJIA), or simply the Dow, comprises 30 prominent companies listed on US stock exchanges. This dataset spans from 2009 to 2024, containing 4035 rows for each stock, detailing the lowest price, highest price, opening price, closing price, and trading volume for each trading day. With historical stock data from the Dow Jones index, encompassing 30 stocks from December 31, 2008, to March 30, 2024, including attributes such as opening price, closing price, maximum price, minimum price, and trading volume for every working day, this comprehensive

dataset forms a solid foundation for training and evaluating our reinforcement learning agents.

B. Performance of Individual Reinforcement Learning Agents

The Table I summarizes the performance metrics for each reinforcement learning agent and the proposed methods. Figure 2 shows the backtest of each agent for cumulative income on training data after training on the same data. As seen, all agents perform well on this data, achieving cumulative income of over 7 times. Although this level of income cannot be considered realistic since the agents were trained on the same data, it can still serve as a benchmark for evaluating the performance of each agent. The TD3 agent has shown relatively better performance compared to the other agents.

Moreover, Figure 3 indicates the backtest of the five agents, as well as the proposed algorithm and the algorithm averaging the weights of all agents, on test data from 2021 to 2024. As shown in the figure, although the TD3 agent performed well on the training data, the DDPG agent performed better on the test data. The algorithm averaging the weights of all agents performed nearly as well as the average of all agents. The proposed algorithm, which uses the average performance of the previous 10 days and attempts to switch to the best-performing agent each day, has not only performed better on average than the weight-averaging algorithm but also outperformed all other agents. As the backtest results show, it achieved a cumulative profit of 38.29%, the best performance among all agents.

Figure 4 illustrates changes in the weight of stocks in the portfolio on each working day. For example, a positive 20% in Apple stock means buying 20% more of this stock than currently in the portfolio. On a working day, a zero percent means holding the same amount of the stock as the previous day. Similarly, a negative 20% means selling 20% of this stock on the current working day. Figure 5, similar to the previous one, shows the average amount of buying, selling, and holding of stocks over a week of working days. The green arrow indicates how much to increase the number of shares in the portfolio, and coming down from hold, which is red, indicates how much to decrease.

Additionally, Table II shows the percentage usage of each agent during the entire backtest period, which includes 765 working days. The proposed method used the DDPG algorithm, which had the best cumulative income, 21.48% of the time, and the PPO algorithm, which had the worst performance, 12.21% of the time.

TABLE I. PERFORMANCE METRICS FOR DIFFERENT REINFORCEMENT LEARNING AGENTS AND PROPOSED METHODS ON THE DOW JONES INDUSTRIAL AVERAGE DATASET (2009-2024).

Evaluation Metrics	A2C Model	PPO Model	DDPG Model	TD3 Model	SAC Model	Proposed Method 1	Main Proposed Method
Annual return	0.10355	0.088963	0.109103	0.093883	0.104561	0.100179	0.114276
Cumulative returns	0.343403	0.2909	0.363759	0.308451	0.347093	0.331145	0.382903
Annual volatility	0.139666	0.139456	0.142429	0.147605	0.13928	0.140483	0.142167
Sharpe ratio	0.775404	0.680907	0.798369	0.681806	0.783763	0.749933	0.832301
Calmar ratio	0.580168	0.578662	0.610732	0.451763	0.59372	0.57494	0.693455
Stability	0.439511	0.521288	0.522191	0.253352	0.505337	0.459214	0.639741
Max drawdown	-0.17848	-0.15374	-0.17864	-0.20781	-0.17611	-0.17424	-0.16479
Omega ratio	1.141502	1.122649	1.144586	1.122642	1.141947	1.135974	1.153198
Sortino ratio	1.117814	0.979952	1.146267	0.975362	1.122939	1.076073	1.20474
Skew	-0.14533	-0.12615	-0.17648	-0.17439	-0.1918	-0.16957	-0.13656
Kurtosis	1.844322	1.724517	1.561056	1.654512	1.784828	1.747927	1.798994
Tail ratio	1.048237	1.024539	1.028713	1.077653	0.969984	0.999069	1.005422
Daily Value at risk	-0.01717	-0.01719	-0.01749	-0.0182	-0.01711	-0.01728	-0.01744
Alpha	0	0	0	0	0	0	0
Beta	1	1	1	1	1	1	1

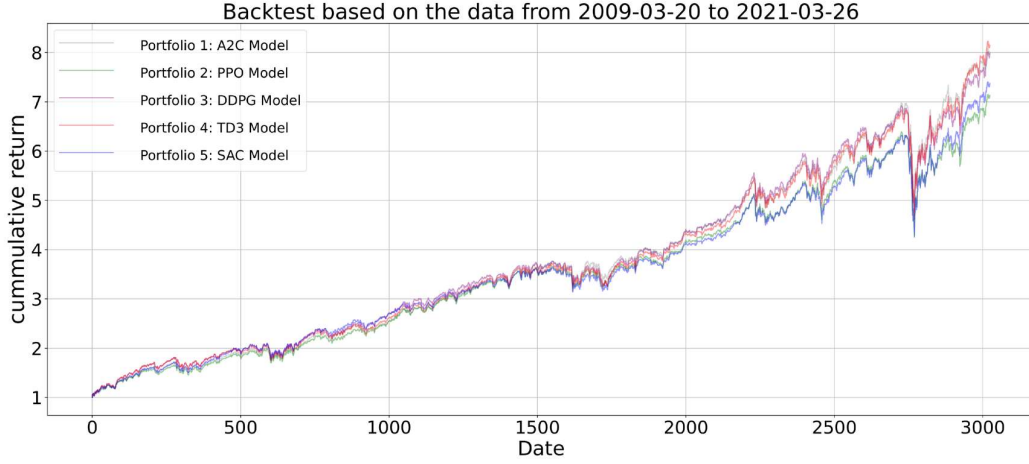


Fig. 2. Backtest on Train Data and Cumulative Return by RL Models.

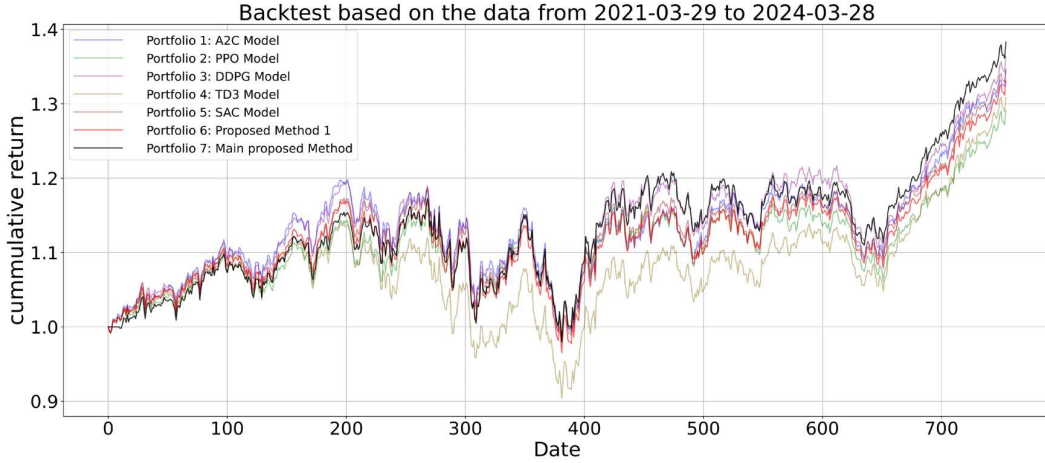


Fig. 3. Backtest on Test Data and Cumulative Return by RL Agents.

TABLE II. THIS TABLE DESCRIBES THE USAGE PERCENTAGES FOR EACH ALGORITHM.

Algorithm	A2C	PPO	DDPG	TD3	SAC
Usage Percent	19.33%	12.21%	21.48%	30.47%	16.51%

V. DISCUSSION

Our study demonstrates the significant potential of deep reinforcement learning (DRL) in optimizing investment portfolios. By dynamically selecting the best-performing DRL agents over short-term horizons, our system adapts to changing market conditions. This adaptability allows it to leverage the strengths of different DRL approaches in real-time, enhancing portfolio performance. Our findings indicate that a multi-agent DRL strategy, compared to a single agent, offers more robust and resilient investment outcomes. Improved risk-adjusted performance metrics, such as higher Sharpe and Sortino ratios, underscore the practical viability of our approach, indicating it boosts returns while effectively managing risk, making it a compelling option for investors seeking growth and stability.

We utilized five reinforcement learning agents: A2C [25], PPO [26], DDPG [27], TD3 [28], and SAC [29] for portfolio optimization. Each agent, due to its specific features, can perform better in different financial market conditions. The advantages of each algorithm, which are summarized in Table III, highlight how they can excel under varying market conditions. After implementation, training with training data, and backtesting the agents using test data, the results according to Table IV for cumulative returns were obtained. The challenge here was to correctly identify the best agent each working day to increase cumulative returns. One solution was to use the weighted average of the portfolio of all agents each working day. As the backtest results show in Table V, this approach led to a cumulative return between the highest (DDPG 36%) and the lowest cumulative return (PPO 29%), amounting to (Proposed Method 1, 33%).

Another solution we employed was to use the agent that had performed best over the past 10 days for each working day. As shown in Table VI, the backtest results indicate an increase in cumulative returns (Main Proposed Method 38%), surpassing both the average cumulative returns of the agents and the best-performing agent.

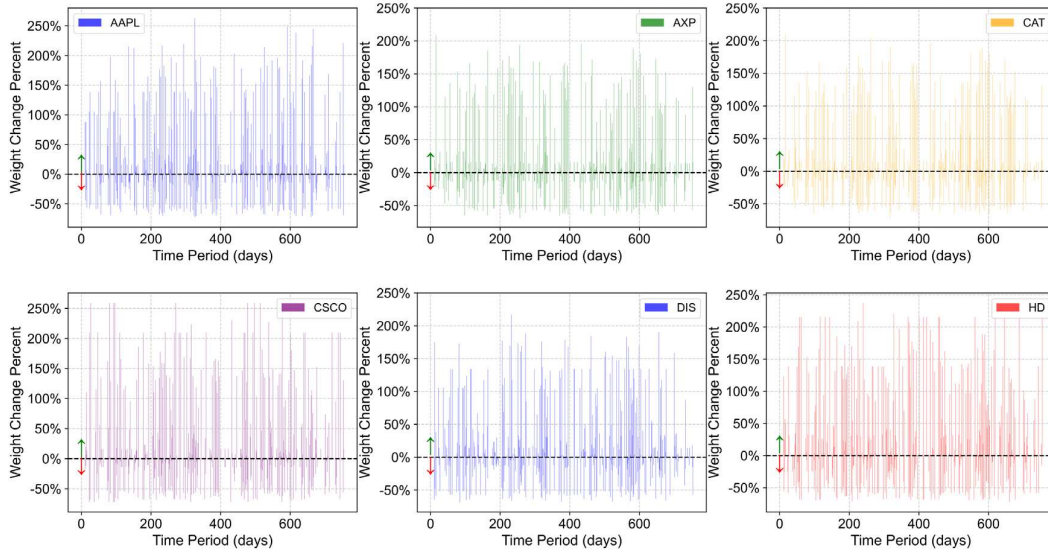


Fig. 4. Daily Portfolio Weight Changes: 6 Example Stocks.

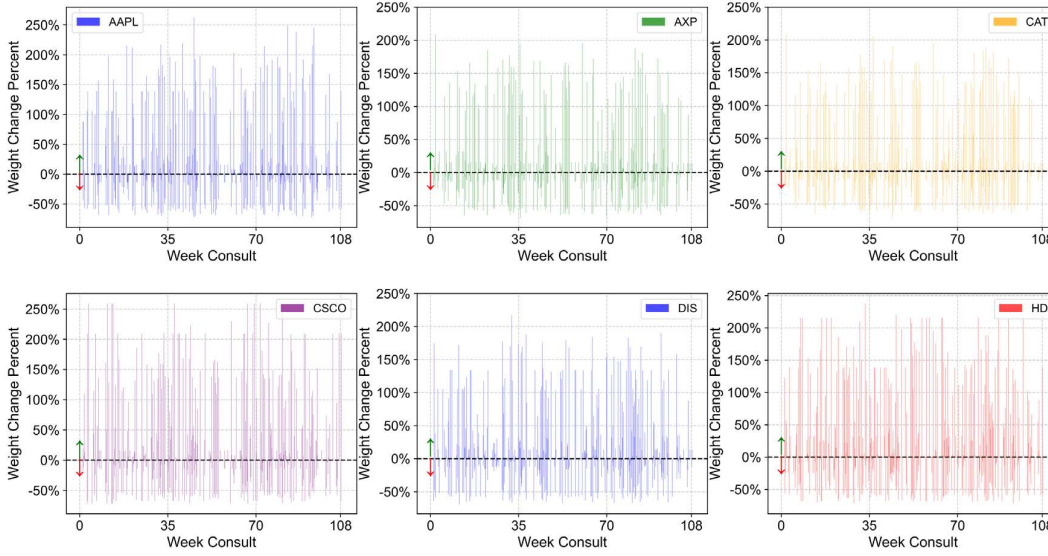


Fig. 5. Weekly Portfolio Weight Changes: 6 Example Stocks.

Our system's success with historical Dow Jones data suggests its potential across diverse markets and asset classes. Further research is essential to validate its performance under varying economic conditions and scalability in real-world settings. Future studies should explore integrating additional DRL agents and advanced decision-making frameworks to enhance portfolio performance. Our findings emphasize the value of dynamic, adaptive strategies in portfolio management, leveraging diverse DRL agents for superior investment outcomes.

TABLE III. ADVANTAGES OF DIFFERENT REINFORCEMENT LEARNING ALGORITHMS FOR PORTFOLIO OPTIMIZATION.

Advantages	Algorithms
Stable, cost-effective, faster, and works better with large batch sizes	A2C
Improve stability, less variance, simple to implement	PPO
Better at handling high-dimensional continuous action spaces	DDPG
Improve network stability in complex environments	TD3
Improve stability	SAC

TABLE IV. CUMULATIVE RETURNS BY MODEL (2021/3/29 – 2024/3/28).

Model	Cumulative Return (%)
PPO Model	29.0%
TD3 Model	30.8%
A2C Model	34.3%
SAC Model	34.7%
DDPG Model	36.3%

TABLE V. CUMULATIVE RETURNS FOR PROPOSED METHOD 1 (2021/3/29 – 2024/3/28).

Model	Cumulative Return (%)
PPO Model	29.0%
TD3 Model	30.8%
Proposed Method 1	33.1%
A2C Model	34.3%
SAC Model	34.7%
DDPG Model	36.3%

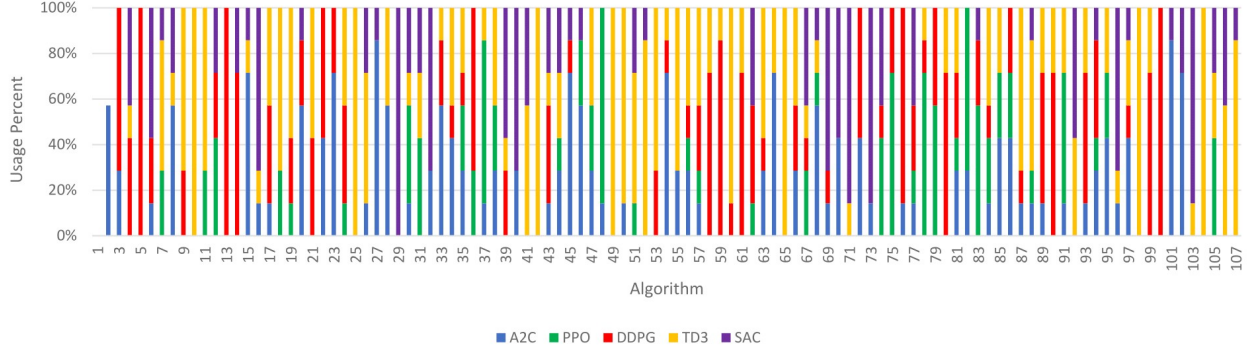


Fig. 6. The Percentage Usage of Agents Throughout Each Working Week.

TABLE VI. CUMULATIVE RETURNS FOR MAIN PROPOSED METHOD (2021/3/29 – 2024/3/28).

Model	Cumulative Return (%)
PPO Model	29.0%
TD3 Model	30.8%
Proposed Method 1	33.1%
A2C Model	34.3%
SAC Model	34.7%
DDPG Model	36.3%
Main Proposed Method	38.2%

VI. CONCLUSION AND FUTURE WORKS

In conclusion, this paper presents a novel consulting system for portfolio asset allocation that harnesses the power of deep reinforcement learning to enhance investment decision-making. By integrating five different DRL agents: A2C, PPO, DDPG, TD3, and SAC, and utilizing a short-term agent selection mechanism based on cumulative returns, our system adapts dynamically to market conditions and significantly improves portfolio performance. The effectiveness of our approach is demonstrated through extensive backtesting on historical Dow Jones index data, achieving superior cumulative returns and risk-adjusted performance metrics compared to individual agents and traditional strategies. These findings underscore the potential of leveraging DRL in advancing portfolio management and highlight the benefits of a diversified, adaptive approach to financial consulting.

Here are several potential future directions to further enhance the proposed portfolio asset allocation system:

- 1) Incorporate additional financial indicators and alternative data sources, such as sentiment analysis from news and social media, to provide more comprehensive market insights.
- 2) Expand the system to include other asset classes like bonds, commodities, and cryptocurrencies for increased versatility.
- 3) Investigate the system's performance in fuzzy environments with highly uncertain and volatile market conditions to validate its robustness and adaptability.
- 4) Explore the potential of transfer learning to adapt models trained on one market to other markets, enhancing cross-market generalization and effectiveness.

- 5) Develop real-time implementation strategies to enable the system to operate efficiently in live trading environments.

VII. CODE AVAILABILITY

The code is available for sharing. Please contact the author to request access.

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